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Temperature and predation alter metabolic scaling without changing the size-based community structure of freshwater macroinvertebrates

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Body size is a key trait that influences ecological processes such as metabolism, abundance and species interactions. While the metabolic theory of ecology (MTE) proposes a universal scaling of metabolic rate with body mass, recent evidence shows that this relationship is not fixed. Environmental factors like temperature and predation can alter the metabolic scaling exponent, potentially reshaping size distributions. However, most research has examined these patterns within individual species, leaving open questions about how environmental drivers affect scaling at the community level. To address this, we performed a mesocosm experiment manipulating both temperature and fish predator presence in freshwater macroinvertebrate communities. We found that warming altered metabolic rates in a predation-dependent way: without fish, metabolic rates increased in large individuals but decreased in small individuals, whereas the opposite occurred with fish present. This suggests that larger individuals reduce their baseline metabolic rates under predation risk, especially at higher temperatures. Interestingly, the slope of the community size distribution remained stable across treatments, indicating that shifts in metabolic scaling occurred independently of changes in size structure. Together, these findings highlight the environmental sensitivity of metabolic scaling and suggest that links between metabolism scaling and size distributions may be more complex than MTE predicts.

This article is part of the theme issue 'Embracing variability in comparative physiology: why it matters and what to do with it'.

1. Introduction

Body size plays a pivotal role in shaping ecological dynamics, influencing key processes such as metabolism, reproduction and species interactions across biological scales [1–3]. A central framework explaining these patterns is the metabolic theory of ecology (MTE), which posits that an organism's metabolic rate (R) scales with body mass (M) following a power–law relationship: $R = aM^b$. The exponent b is traditionally thought to approximate three-quarters, a value theoretically derived from models describing internal resource

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transport networks [4]. This scaling rule has been widely used to predict how energy flows through individuals, populations and entire ecosystems [1,5].

Nonetheless, empirical work has increasingly shown that the exponent b is far from universal. It often varies across taxa, environments and life stages [6–10]. In aquatic macroinvertebrates, for instance, metabolic scaling can shift in response to both temperature and predation pressure, often becoming shallower than the canonical three-quarters expectation [9,11,12]. These observations suggest that ecological conditions can significantly influence metabolic allometry, revealing flexibility in a phenomenon once considered physically constrained.

Variability in metabolic scaling has potentially important implications for the individual size distribution (ISD)—a well-established pattern where smaller organisms tend to occur at higher densities than larger ones [13,14]. Assuming energy use is equal across size classes, shifts in b should map directly onto changes in size spectra under equilibrium conditions [7,15,16]. The ISD is described by a power law, $N \sim M^{\lambda}$, where N is abundance and M is individual body mass. Metabolic scaling theory predicts that the exponent λ is generated by interactions of three community-level variables: trophic transfer efficiency (TTE), predator–prey mass ratio (PPMR) and metabolic scaling (b) [17–20], such that:

$$\lambda + 1 = \frac{\log_{10} PPMR}{\log_{10} TTE} - b\,. \tag{1.1}$$

Given typical values of $PPRM$ as 10^4, of TTE as 0.1 and of b as 0.75, the exponent λ tends to be negative, with values typically approximately −2 to −1.75, underscoring a remarkably consistent ecological pattern across Earth's diverse ecosystems [21,22]. Yet, the link between the value of the metabolic scaling exponent b and the ISD's λ has been rarely studied experimentally. Recent work demonstrated that even within a single species, evolutionary changes in body size can decouple metabolic scaling from abundance patterns, underscoring how energetics and demography interact in more nuanced ways than previously assumed [16].

Environmental temperature adds further complexity to metabolic scaling by differentially affecting metabolic rates across body sizes. Empirical evidence indicates that warming can alter intraspecific metabolic scaling, often leading to size-dependent metabolic responses in which smaller individuals remain more thermally responsive than larger ones. Such patterns have been documented, for example, in amphipods [9,12] and are consistent with predictions of the metabolic-level boundaries hypothesis, which proposes that warming may reduce the scaling exponent b through shifts in the relative contributions of surface-area- versus volume-related metabolic processes. Importantly, these temperature-driven changes in metabolic scaling can occur independently of predation pressure or behavioural suppression and may modify the mass–metabolism relationship without necessarily producing immediate changes in population or community size structure. Rising temperatures not only elevate metabolic demand through basic biochemical processes [23], at least up to a point before decreasing [24], but also they can shift the body-mass scaling relationship itself via size-specific physiological and behavioural responses [8,9,12]. Moreover, temperature influences body size more directly through well-documented plastic responses, such as the temperature–size rule [25–27]. Warmer environments are predicted to favour smaller organisms [28] leading to more negative values of λ (i.e. 'steeper' slopes), and this has been observed empirically [27,29–32]. Consequently, these size reductions may influence λ, linking thermal conditions to both metabolic scaling and ISD [16].

Predation, especially when it targets larger individuals, adds an important layer of ecological pressure. In many systems, larger prey face a greater risk of being eaten, often driving populations and communities towards smaller average body sizes [33,34]. These changes go beyond simple shifts in size distributions; they also influence the way metabolic rates scale with body mass, by favouring traits that lower energy demands, such as reduced activity and slower growth in larger individuals. While the metabolic rates we measured in previous studies reflect resting conditions, they are tightly connected to an individual's overall capacity for movement, growth and other energy-intensive functions and are sensitive to ecological pressures. Studies of amphipods have shown that exposure to higher temperatures and predator presence triggers size-specific adjustments: larger individuals tend to suppress energetically costly behaviours to avoid detection, while smaller, less visible individuals remain more metabolically responsive [9,12]. This suggests a clear adaptive pattern. In environments with active predators, large individuals often restrict activity and growth—especially at higher temperatures when encounters are more frequent [35]—leading to a dampened metabolic response among adults. Meanwhile, smaller individuals, under less threat, continue to ramp up metabolic activity with warming, producing a shallower metabolic scaling (b). In predator-free settings, this pattern flips: small individuals appear to suppress metabolic responses, possibly due to risk of cannibalism from larger conspecific individuals, while adults take advantage of safer conditions to increase their metabolic rates. This reversal causes b to rise with temperature, a trend particularly evident in amphipods [9,12,36].

Although the effects of temperature and predation on metabolic scaling have been explored extensively, these investigations have largely focused on single species or isolated populations. What remains unclear is how these two drivers interact to influence metabolic scaling at the community level, where species vary in body size, life history and ecological roles. Crucially, there is still a lack of experimental studies that directly examine how changes in metabolic scaling correspond to shifts in ISD under combined thermal and predation pressures. This missing link limits our understanding of how physiological processes scale up to influence community structure [16,37]. As a result, we still lack empirical evidence about whether and how environmentally driven changes in metabolism propagate to reshape ISD at the community scale. Addressing this gap is essential to understand how temperature and predation jointly structure energy flow and body size patterns in natural systems.

To address this gap, we conducted a mesocosm experiment manipulating temperature and predator presence in freshwater macroinvertebrate communities. Our aim was to evaluate how these environmental drivers affect community-level metabolic scaling exponents and in turn, community body-size distributions. Specifically, we tested: (i) how does ecological context influence the metabolic scaling exponent b at the community scale and (ii) do changes in b predict shifts in the ISD's λ.

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By linking individual-level metabolism, which underlies community-level metabolic scaling, with body size distributions that shape size spectra, our work provides a valuable test of how physiological and structural patterns may emerge at the community scale. This approach offers a more ecologically grounded perspective on the scaling of metabolism and population abundance under environmental change.

2. Material and methods

(a) Study site

The experiment was conducted at the University of South Dakota's Experimental Aquatic Research Site, Vermillion, SD. The site consists of 24 fibreglass tanks (1136 l). The tanks were filled with 714 l of water over a mix of cobble and sand for substrate in May of 2023. Water levels were monitored and maintained over the course of the experiment. Each tank had an overflow spout and a magnetic drive water pump (Danner Supreme Aqua-Mag) to ensure mixing and to bubble the surface to maintain sufficient O_2. After filling, tanks were inoculated with approximately 100 ml of filtered (250 µm) local river water to introduce plankton. Populations of insects colonized the tanks via oviposition beginning approximately 1 day after they were filled and continuing through the experiment. AQQA Aquarium 800W Heaters (AQQA Inc.) were placed in half of the tanks and set to approximately 5°C above ambient nighttime temperatures. Temperature data were taken daily in the middle of the water column using a YSI Pro-Series DSS (YSI Inc., Yellow Springs, OH, USA). Heaters were adjusted to maintain a temperature of approximately 5°C above ambient (electronic supplementary material, figure S1). When averaged across dates, the heated mesocosms were warmer than the unheated mesocosms by 4.3 ± 1.9°C. Fish (*Lepomis cyanellus*) were gathered from a local pond with a seine net and set free in half of the tanks on 10 May 2023, 6 days after filling the tanks (one fish per tank). At the beginning of the experiment, their lengths ranged from roughly 8 to 10 cm, and by the end, they had grown to between 10 and 12.5 cm. The setup included four treatment combinations: ambient temperature with fish, ambient temperature without fish, heated tanks with fish and heated tanks without fish (figure 1).

(b) Macroinvertebrates

The 24 mesocosm tanks supported naturally assembling freshwater invertebrate communities. These communities developed from colonization by adult aquatic insects that used the tanks to lay their eggs, resulting in the emergence of a diverse and self-assembling community. On average, each tank contained around seven taxa, with most taxa consistently present across all tanks. The communities included a range of freshwater invertebrates such as midges, beetles and various odonate larvae (e.g. damselflies and dragonflies), several of which are known aquatic predators.

To investigate patterns in ISD, we measured a total of 1529 individuals across these mesocosms. Tanks with fewer than 100 individuals were excluded from the ISD analysis to ensure sufficient sample size for reliable estimation of ISD slopes, as estimates of power–law parameters are known to be unstable at low sample sizes. The three excluded tanks were distributed across treatments and were not all predator or heated treatments, indicating that exclusion reflected low overall colonization rather than systematic treatment effects. For the metabolic scaling analysis, 415 individuals were measured, with approximately 50 individuals sampled from each of three replicate tanks per treatment. Sampling was designed to include all species present within each tank and to capture a representative range of body sizes. This ensured robust comparisons of metabolic scaling across treatments, accounting for the diversity and size structure of the invertebrate communities.

(c) Metabolic rate measurements

Metabolic measurements occurred 30 days after treatments were started. A Hess sampler (0.032 m^2, 500 µm collection mesh) was used to sample benthic macroinvertebrates following the methods of Forster *et al.* [26]. Macroinvertebrates from the sediment were sorted and placed in containers with filtered water (Whatman GF/F 0.7 µm glass fibre filters to remove metabolically active microbes) from their corresponding treatment and held without food for 24 h to empty their guts. After 24 h, they were separated into individual glass vials with 20 ml of oxygenated, filtered water (electronic supplementary material, table S2). Nine experimental vials with a macroinvertebrate and one control without (to correct for oxygen changes not due to respiration) were used for each run of measurements. Oxygen concentrations were measured using a fibre-optic oxygen sensor Fibox 4 oxygen sensor (Presens, Regensburg, Germany) that allowed readings to be taken externally through the vial wall without opening, moving or agitating the chambers. Vials remained stationary throughout the incubation period, and animals were not re-handled or disturbed during oxygen measurements, minimizing the potential for activity-related inflation of metabolic rates [38]. The macroinvertebrates were then photographed (lengths measured using ImageJ) before being dried and measured on a microbalance. Oxygen consumption was estimated using the following equation:

$$R = \frac{(PE - PC)S \times A \times V}{t}.$$

R is the oxygen consumption, PE and PC represent partial pressures of oxygen in the experimental environment (20 ml vial with an invertebrate) and control, respectively, S is the solubility coefficient of oxygen in water, A is the volume of 1 mol O_2 at standard temperature and pressure, V is the volume of water in the vials and t is the time of incubation.

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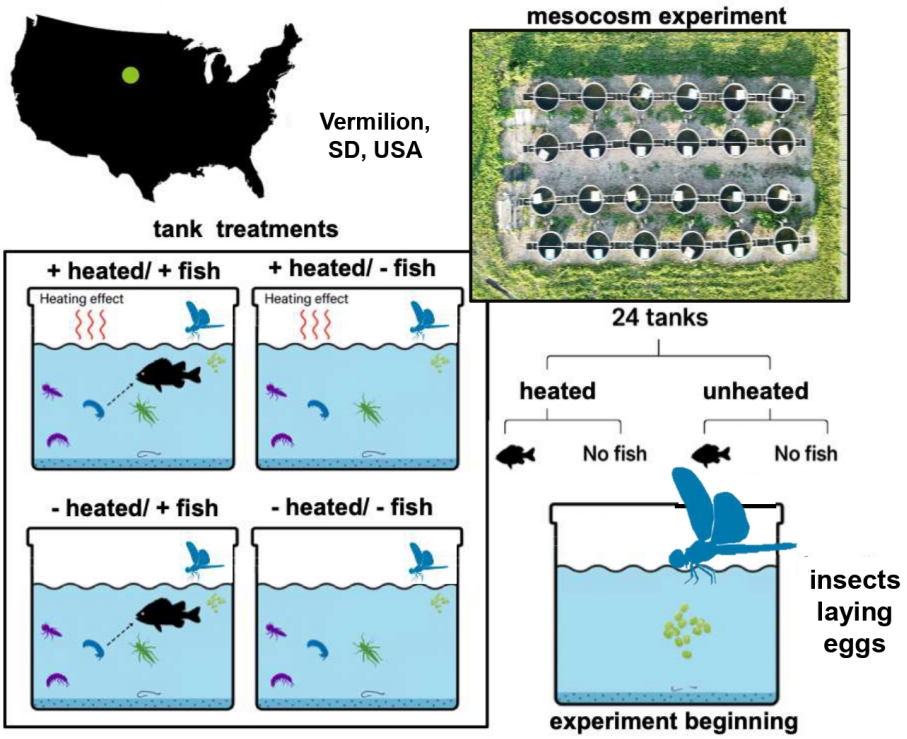


Figure 1. Overview of the experimental design. The study was conducted in Vermillion, South Dakota, USA (green dot on map), using 24 outdoor field mesocosms arranged in a 4 × 6 grid (aerial image, top right). Tanks were assigned to a fully factorial design crossing two treatments: temperature (heated versus unheated) and fish presence (with versus without *Lepomis cyanellus*), resulting in four treatment combinations. Schematics depict each treatment combination, with red wavy arrows indicating heated conditions and silhouettes of *L. cyanellus* representing fish presence. Macroinvertebrate prey of varying sizes and taxa naturally colonized the tanks. At the start of the experiment, aquatic insects oviposited into the tanks (bottom right), initiating community assembly and continued to oviposit throughout the experiment. This design allowed us to test the interactive effects of temperature and predation on community-level metabolic scaling and size spectra.

To ensure accurate results, preliminary testing was performed to see how long it would take for a visible drop in dissolved oxygen content to occur. This determined the 60 min time span between measurements. During the experiment, macroinvertebrates were kept in water that corresponded to their respective thermal treatment of predation and temperature. In addition, to minimize handling effects and ensure estimation of resting metabolic rate, after transfer into individual respirometry vials, individuals were allowed to settle undisturbed in still, oxygenated water before oxygen measurements were taken. Preliminary trials indicated that oxygen concentration declined linearly over a 1-h incubation period, suggesting stable metabolic rates during measurements. Although recovery dynamics can vary among taxa, all individuals were handled using identical protocols, and measurements were conducted under low-light, low-disturbance conditions. This standardized approach is commonly used in comparative respirometry of aquatic invertebrates and is appropriate for assessing relative metabolic rates and scaling relationships across species and treatments.

(d) Individual size distribution

On the last day of the experiment (after 30 days), we collected benthic macroinvertebrates with a Hess sampler. The sampler was forced through the sediment and rock substrate so it was flush with the bottom of the mesocosm. Then we disturbed the substrate by hand for 10 s and flushed the substrate through a 250 µm mesh net [39,40]. The sample was then placed on a 250 µm sieve, large debris (e.g. rocks) was scrubbed and removed, and the remaining sample was preserved in 95% EtOH. To measure body sizes of benthic macroinvertebrates, we sorted a subsample and retained the first 200–500 individuals. This approximates the number of individuals in simulation studies for which estimates of ISD stabilize [40]. These individuals were placed on a gridded watch glass, photographed and measured for length to the nearest pixel using ImageJ. The pixels were converted to mm using the grid as a reference and adjusting for changes in magnification. Then the dry mass (mg) of each individual was determined using family-specific length–mass equation [41].

(e) Data analysis

To estimate the community-level metabolic scaling exponent b, we fitted log-transformed respiration as a function of log-transformed body mass using a Gaussian linear mixed model. Each individual's respiration rate was measured and matched to its body mass, and these individual-level data were nested within mesocosm tanks, which served as the unit of replication. The model included a three-way interaction among log body mass, temperature and predator treatment, with tank identity as a varying intercept. This structure allowed us to estimate tank-level scaling relationships while accounting for differences in intercepts across replicates. The prior for the scaling exponent was set to *Normal* (0.75,0.2) to reflect prior assumptions of

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the three-quarters scaling pattern. The remaining regression coefficients were set to *Normal* (0,1), and the intercept was *Normal* (0,0.2). Both log respiration and log mass were mean-centred prior to analysis to facilitate model convergence and align priors with the centred data. The resulting community-level value of b thus reflects the average slope across tanks, while variation in intercepts captures differences in baseline metabolic rates across communities.

To estimate ISD's λ, we fitted a generalized linear mixed model with individual body size as the response variable, heat and fish treatment and their interactions as the predictor variables, and tank as the varying intercept. The likelihood was a truncated Pareto, which allows the model to estimate coefficients reflecting the frequency distribution of the individual body sizes λ [42]. Additional information required for the truncated Pareto is $x_{\min}$ and $x_{\max}$, which represent the minimum and maximum body sizes from each tank. To set $x_{\min}$, we used the *estimate_xmin()* function from the *poweRlaw* package, which determines the minimum body size for which the data still follow a power law. For $x_{\max}$, we used the measured maximum body size in each tank. Importantly, fish as organisms were excluded from all analyses of size distribution; only invertebrate body sizes were used to estimate λ.

We fitted the models using *brms* [43] and *isdbayes* [42] in R v. 4.4.2 (R Core Team 2024). Posteriors were explored with the NUTS algorithm in Stan (Stan Development Team 2024). We used 2000 iterations and 4 chains for each model. Model convergence was checked via plotting the chains and ensuring that R-hat was less than 1.01. Model fit was checked using posterior predictive checking [44] (electronic supplementary material, figures S2 and S3). Prior implications were checked using prior predictive checking [45] (electronic supplementary material, figure S4).

(f) Simulation study

Our experimental results revealed pronounced treatment effects on community-level metabolic scaling b, but little corresponding change in the ISD slope λ. To help interpret this apparent decoupling, we conducted a simulation study grounded in metabolic scaling theory. The purpose of this simulation was not to reproduce the experimental results directly, but to explore how variation in key food-web parameters, including metabolic scaling b, TTE and PPMR, can influence λ. By examining λ across a broad range of plausible parameter combinations, this analysis provides insight into whether compensatory changes among these drivers could plausibly explain the observed stability of λ across temperature and predation treatments. To determine how likely our empirical estimates of λ and b met the assumption of the MTE, we used equation (1.1) to simulate λ across a range of plausible values of *PPMR* (0.05–0.4), *TTE* (10^2–10^7) and b (0.3, 0.75, 1) and compared those values to empirical λs from the mesocosms.

3. Results

(a) Insect collections

We collected a total of 6320 benthic insects. Of these, 5906 were used for the ISD and 414 for metabolic scaling. Chironomidae made up 97% of abundance (i.e. 6155/6320) and 85% of biomass. Those proportions of chironomids were similar in the ISD and metabolic scaling data sets for abundance (91 and 97%, respectively) and biomass (75 and 85%, respectively). The remaining common taxa were Ceratopogonidae, Odonata and Coleoptera. All other taxa were present in only one or two mesocosms and made up less than 1% of abundance and less than 0.01% of biomass. Across treatments, abundance was similar with Chironomidae making up 92–99% of individuals. However, biomass varied somewhat by heat treatment, with Chironomidae making up 77% of biomass in the heated treatment and 94% in the unheated treatments. The difference was driven by higher relative biomass of Odonata in the heated (14%) versus unheated (2%) treatments (table 1).

(b) Results summary

Our results support the hypothesis that warming increases the community-level metabolic scaling exponent when predator risk is absent, but that this effect is reversed under predation pressure. However, the lack of change in λ across treatments suggests that shifts in individual-level energetics may not directly translate into changes in community-level size structure, indicating a possible disconnect between metabolic scaling and emergent ISD patterns.

(c) Metabolic scaling

Metabolic scaling exponents responded to both heating and predation treatments (table 2, figure 2a). In the presence of fish predators, heating caused a clear reduction in the metabolic scaling exponent, from 0.46 under ambient conditions to 0.35 under heated conditions (table 2), indicating a greater than 99% probability of decline. This pattern is consistent with a suppression of metabolic rate in larger individuals, as previously hypothesized in predator-sensitive taxa. In contrast, when fish were absent, heating increased the metabolic scaling exponent substantially, from 0.31 at ambient temperature to 0.45 under heated conditions (table 2), indicating a greater than 99% probability of an increase. These opposing trends demonstrate a strong interactive effect of temperature and predation on the community-level metabolic scaling relationship.

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Table 1. The relative abundance and biomass of the two most common taxa (Chironomidae and Odonata) in the benthic insect community. Data show percentage abundance and biomass of each taxon per mesocosm relative to the total insect biomass collected in a mesocosm. Mesocosm numbers show the mesocosm order in our experimental array from first (1) to last (24). They are scattered here to emphasize the random assignment of treatments among the 24 mesocosms.

		abundance (%)		biomass (%)	
mesocosm	treatment	Chironomidae	Odonata	Chironomidae	Odonata
1	+fish|+heat	96	4	66	34
4	+fish|+heat	99	1	61	39
9	+fish|+heat	82	1	59	7
15	+fish|+heat	99	1	100	0
18	+fish|+heat	67	11	64	5
19	+fish|+heat	100	0	100	0
5	+fish|−heat	98	0	94	0
7	+fish|−heat	100	0	100	0
11	+fish|−heat	99	0	91	0
14	+fish|−heat	97	0	94	0
20	+fish|−heat	100	0	100	0
24	+fish|−heat	100	0	100	0
3	−fish|+heat	100	0	100	0
10	−fish|+heat	95	1	51	15
12	−fish|+heat	93	1	94	3
17	−fish|+heat	99	1	100	0
21	−fish|+heat	100	0	100	0
23	−fish|+heat	99	1	61	39
2	−fish|+heat	100	0	100	0
6	−fish|−heat	92	2	79	13
8	−fish|−heat	99	0	99	0
13	−fish|-heat	97	0	89	1
16	−fish|-heat	99	1	90	10
22	−fish|-heat	100	0	100	0

(d) Individual size distributions

The slope of the ISD's λ remained largely consistent across both temperature and predation treatments (figure 2b, table 2). Across treatments, posterior estimates of the ISD slope (λ) ranged narrowly from –1.63 (median) in the ambient fish treatment to –1.56 in the heated treatment without fish, but with overlapping credible intervals indicating no meaningful treatment effects (table 2). Specifically, neither warming nor the presence of fish predators altered the value of λ. This suggests that, despite changes in community temperature and predator cues, the overall structure of abundance relative to body size remained stable. Thus, while temperature and predation clearly influenced community-level metabolic scaling (see below), these effects did not directly translate into detectable shifts in community-level size–abundance patterns.

(e) Simulation study

Across individual mesocosms, λs ranged from −2.5 to −1.1 (posterior medians). To place this range into context, we compared it to λs simulated using equation (1.1). Figure 3 shows λ values derived from simulated and empirically observed size distributions across a wide range of food-web parameter combinations. The broader spread of λ values in this figure reflects variation in simulated TTE, PPMR and metabolic scaling (b), rather than uncertainty or variability in the posterior treatment-level estimates. Equation (1.1) indicates that λ is shaped by the interplay among *TTE*, *PPMR* and the metabolic scaling exponent b. However, variation in these parameters does not necessarily lead to shifts in λ. Simulations from equation (1.1) across a wide range of *TTE* and *PPMR* values show that our empirical λs are most plausible when b is less than 0.75 (figure 3). Notably, empirical λ values from our study overlapped most closely with the simulated λ values when the metabolic scaling exponent b was less than 0.75 (figure 3). This pattern is evident because, in the panels where b ranged between 0.3 and 0.5, the cloud of simulated λ values generated by combinations of *PPMR* and ecologically plausible *TTE* (dashed horizontal lines) aligns more closely with the vertical lines representing empirical λ estimates. This suggests that the observed invariance in λ across environmental

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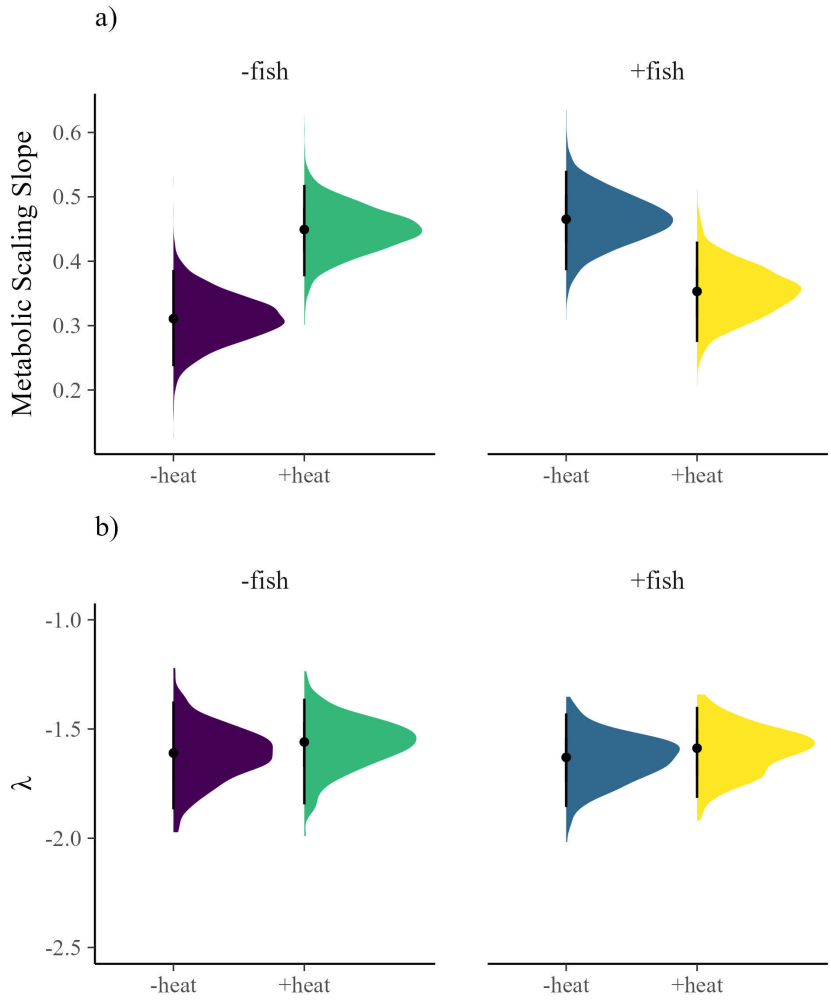


Figure 2. Horizontal density plots show the full posterior distributions of (a) the metabolic scaling slope and (b) the ISD slope (λ) across four treatments in a factorial experiment (ambient versus heated temperature, crossed with presence versus absence of fish predators). Points represent posterior means, and error bars show 95% credible intervals. Colour indicates treatment group. While heating did not significantly affect λ (b), it did alter the metabolic scaling b (a), with contrasting effects depending on predation regime. Specifically, heating reduced the metabolic scaling slope in the presence of fish but increased it in their absence, indicating interactive effects of temperature and predation on metabolic allometry, independent of changes in community size structure.

Table 2. Median and 95% credible intervals for the metabolic scaling slope b and the size-abundance parameter λ estimated from the individual size distribution (ISD).

treatment	response	median	low_{95}	$high_{95}$
−heat|+fish	metabolic scaling slope b	0.46	0.4	0.53
−heat|−fish	metabolic scaling slope b	0.31	0.25	0.38
+heat|+fish	metabolic scaling slope b	0.35	0.28	0.42
+heat|−fish	metabolic scaling slope b	0.45	0.39	0.51
−heat|+fish	ISD λ	−1.63	−1.86	−1.43
−heat|−fish	ISD λ	−1.61	−1.87	−1.37
+heat|+fish	ISD λ	−1.59	−1.82	−1.4
+heat|−fish	ISD λ	−1.56	−1.84	−1.36

treatments could be explained by compensatory shifts in food-web parameters that buffer the effects of metabolic scaling changes.

4. Discussion

This study contributes to growing evidence that metabolic scaling is not fixed, as proposed by the MTE [1], but instead responds flexibly to environmental conditions [9,11,12]. We found that temperature and predation exert interactive effects on metabolism, highlighting the importance of ecological context in shaping physiological traits. Specifically, in the absence of fish predators, warming increased the metabolic rates of relatively large macroinvertebrate individuals but decreased the metabolic

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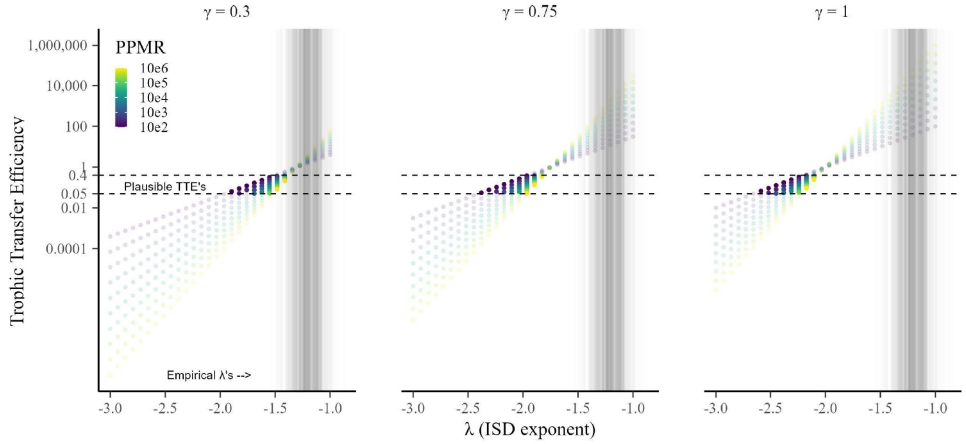


Figure 3. Simulated effects of metabolic scaling exponent *b*, trophic transfer efficiency (TTE), and predator–prey mass ratio (PPMR) on the ISD slope (λ). Panels show combinations of *λ* values across different *b* values (0.3, 0.75 and 1). Each coloured point represents a unique pairing of *TTE* and *PPMR*, with colour indicating *PPMR* magnitude. Horizontal dashed lines mark the range of ecologically plausible *TTE* values, while vertical grey lines indicate empirically observed *λ* values. The figure demonstrates that multiple parameter combinations can yield similar *λ* outcomes focusing on the *y*-axis from approximately 0.01 to 100, suggesting that *λ* can remain invariant even when underlying metabolic and trophic parameters shift. Note that *λ* values shown here reflect simulated and observed size-distribution outcomes across parameter space and are not posterior treatment-level estimates from the Bayesian model summarized in table 2.

rates of smaller individuals. By contrast, the opposite occurred in the presence of predators, thus in the absence of predators increased (consistent with reduced metabolic rates of smaller individuals), but dampened metabolic responses when predators were present increased (consistent with reduced metabolic rates of larger individuals), indicating a strong interaction between thermal environment and predation pressure.

Previous work has shown that predator cues can modify the temperature dependence of metabolic scaling through size-dependent behavioural responses, with larger individuals often suppressing activity under predation risk while smaller individuals remain more metabolically responsive [9,12]. Size-selective predation can favour behavioural suppression in larger, more vulnerable individuals, leading to reduced activity and lower temperature-driven increases in metabolic rate. A similar mechanism may be operating in our mesocosms. In the presence of fish predators, larger more conspicuous macroinvertebrates may have minimized their exposure by dampening an increase in their activity and associated metabolic rate in response to increasing temperature, whereas smaller individuals exhibited stronger thermal responses, resulting in a reduced community-level metabolic scaling exponent *b* [12]. In contrast, in predator-free mesocosms, larger invertebrates may have experienced relaxed top-down control and higher activity levels, producing a stronger metabolic response to temperature at larger body sizes. Together, these patterns suggest that the interactive effects of temperature and predation on metabolic scaling observed here reflect emergent, community-level responses mediated by size-dependent behaviour rather than species-specific mechanisms.

Macroinvertebrate communities in the tanks included individuals ranging from approximately 1 to 15 mm in body length, spanning multiple taxonomic groups (e.g. midges, beetles, damselflies, dragonflies). This wide range of body sizes emphasizes that both small, cryptic taxa and larger, conspicuous individuals were present and likely contributed to the observed size-dependent responses. However, in the absence of fish predators, the large, predatory macroinvertebrates may have had higher activity levels that posed a greater predation risk to small macroinvertebrates, leading to the small invertebrates having reduced activity levels and metabolic rates. Although the overall distribution of macroinvertebrate body sizes did not differ significantly among treatments, it is possible that large predatory invertebrates were more active or behaviourally dominant in fish-free environments [40].

These observations support a broader conceptual framework in which mortality risk—from either predators or conspecifics—can significantly influence metabolic scaling patterns [46]. They also highlight the complexity of predicting physiological outcomes in multitrophic systems, where responses are shaped not only by abiotic conditions but also by the structure and behaviour of interacting species. A mechanistic understanding of ecological scaling must therefore account for both environmental variation and indirect effects mediated through trophic interactions. At the community level, our results contribute to a limited but growing body of research exploring how size scaling patterns manifest across multiple interacting species. This perspective is critical for predicting how entire ecosystems will respond to climate change, as many core processes—such as energy flow and trophic efficiency—emerge from species interactions rather than strictly individual physiology.

Within this context, our respirometry approach was designed to support comparative, community-level inference rather than to estimate absolute baseline metabolic rates for individual species. Although intermittent or continuous flow respirometry conducted over extended timeframes represents the gold standard for estimating standard metabolic rate, such approaches are often impractical for studies involving large numbers of small-bodied organisms spanning multiple taxa. We therefore used short-term closed-vial respirometry, which likely reflects resting or near-resting metabolic rates rather than absolute standard metabolic rates [38]. Importantly, all individuals were measured using identical handling, acclimation, and measurement protocols, allowing robust comparisons of relative metabolic rates and metabolic scaling relationships across temperature and predation treatments. Consequently, our conclusions focus on treatment-induced differences in metabolic scaling at the

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community level, rather than on absolute metabolic values of individual taxa. In our experiment, we applied a community-level approach by examining the ISD slope (λ) as a proxy for energy distribution across body sizes. Unexpectedly, λ remained stable across treatments, despite clear treatment effects on metabolic scaling. This result is surprising because one might expect fish predation to reduce the number of large macroinvertebrates, thereby decreasing λ. Although changes in individual metabolic rates—such as increased respiration under warming or the removal of larger individuals by predators—should cascade to influence ISD, we did not observe this.

Several hypotheses may help explain this apparent inconsistency. First, fish predation may not have been strongly size selective. Although sunfish (*Lepomis* spp.) are known to preferentially consume larger macroinvertebrate prey [47,48], prey selectivity is often size-dependent and can vary with the size or developmental stage of the fish. Smaller sunfish, limited by gape size, tend to consume smaller prey, while larger individuals are capable of targeting larger and a wider range of prey items [49]. A lack of strong size-selectivity would also influence our interpretation of the temperature-related changes in metabolic scaling, which in other studies have been attributed in part to predator-induced reductions in growth or activity in larger individuals [8,12].

Second, an explanation for the consistent λ values across treatments may lie in the balance among key ecological parameters that shape ISDs such as *TTE*, *PPMR* and the metabolic scaling b. While these factors may shift in response to environmental conditions like temperature and predation, their combined influence on energy flow could remain stable. This idea is supported by previous findings from marine systems, where neither *PPMR* nor *TTE* showed consistent variation across thermal gradients [50]. Across a range of b values—from 0.3 to 1—combinations of *PPMR* and *TTE* often yield λ values within empirically observed ranges. In particular, plausible *TTE* values (around 0.05–0.15) align with ISD λ values between –1.5 and –2.5, regardless of b. This suggests that even if metabolic scaling varies with temperature or predator exposure, the broader structure of energy distribution across body sizes can remain unchanged. Such stability could result from compensatory adjustments. For instance, warming tends to reduce the biomass of larger consumers [51], which would tend to lower PPMR. At the same time, it can reduce TTE by decreasing the nutritional quality of prey [52]. If the relative changes in *PPMR* and *TTE* are proportionate, their ratio—and therefore λ—can remain stable. However, unequal changes between the two can lead to steeper λ values by restricting energy transfer to higher trophic levels [53]. Additionally, predator feeding behaviour may further influence *PPMR* under warming. If omnivorous predators like sunfish become more flexible in their foraging—shifting towards smaller or suboptimal prey—this can further lower realized PPMR, buffering λ against steeper declines even as top-down control weakens. Omnivory, in particular, has been shown to compress food-chain length and reduce *PPMR* by broadening the trophic niche towards smaller prey sizes. What this suggests is that even when b varies under different environmental treatments, energy flow across the system may remain effectively unchanged. As a result, size distribution structure at the community level can remain stable, even as community-level metabolism shifts in response to temperature and predation pressure.

Third, the absence of variation in λ across treatments points to the need for a deeper understanding of how temperature and predation shape iISDs, particularly over short timescales. One consideration is that the 30 day exposure to predators may not have been long enough for top-down forces to produce detectable changes in community structure. While metabolic responses to temperature and predation risk can occur quickly [9], changes in population-level patterns—such as abundance distributions—often develop more gradually through processes like survival, growth and reproduction [33,54]. In this context, the stability in λ may reflect a delay in demographic responses rather than a true absence of ecological effect. While the temperature treatment may have underestimated the effect of warming on community structure, evidence from the experiment confirms that both temperature and predation treatments were ecologically active, as fish exhibited clear increases in body size, nearly doubling their mass (electronic supplementary material, table S1), and heated tanks consistently maintained elevated temperatures relative to ambient conditions (electronic supplementary material, figure S1).

Fourth, predator presence may have influenced species colonization or composition rather than across species size structure. Fish may deter or exclude certain taxa from colonizing or persisting in enclosures, not because of prey size *per se*, but due to species-specific behavioural or ecological traits. For example, fish-tolerant species may have broader size ranges or faster life histories that buffer the community against structural change. In this scenario, community composition could shift in response to predation without producing a change in λ, if the size distributions of new and excluded species were functionally redundant. Such species sorting processes have been observed in other aquatic systems, where predators alter community composition without systematically truncating size spectra [55,56].

Taken together, these hypotheses underscore the complexity of scaling relationships in ecological systems. While individual-level metabolic traits clearly respond to environmental variation, these responses do not necessarily translate to predictable changes in community structure. This finding adds to growing evidence that the MTE's assumptions—particularly regarding the tight coupling of metabolism, body size and abundance—are overly simplistic [37,57]. This implies that community-level structure may be buffered against short-term physiological changes, perhaps due to compensatory shifts in species composition, colonization or redundancy in size distribution (figure 4). Such resilience highlights the need to link individual energetics with higher-order ecological structure to fully understand how ecosystems respond to environmental changes.

A critical insight from this study is that macroinvertebrates colonized the mesocosms naturally, resulting in a self-assembling, dynamic community more representative of real ecosystems than artificially assembled ones. This continuous recruitment, combined with natural community cycling and turnover, likely played a key role in buffering the system against structural disruptions from both warming and predation. In such open systems, ecological interactions—such as dispersal, recolonization and species sorting—can maintain community composition and size structure even under substantial environmental stress [58–60]. Because community structure in these systems emerges from the collective dynamics of multiple taxa rather than from the dominance of any single species, our analyses were explicitly conducted at the community level, integrating individuals across taxa and size classes within each mesocosm. Although species differ in metabolic traits, thermal tolerance

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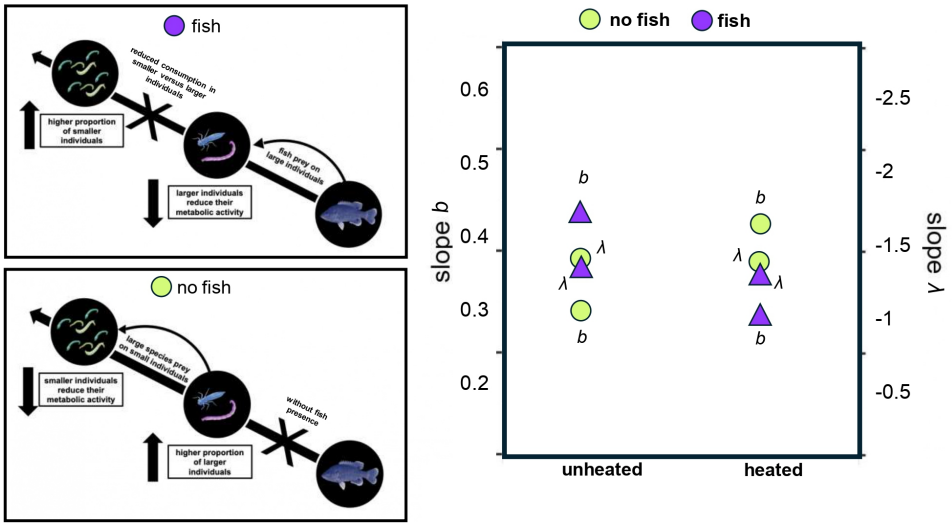


Figure 4. Conceptual diagrams of our hypotheses (left) and empirical results (right) showing how fish predation and temperature influence size structure and metabolic scaling. In systems with fish (top left), we predicted that predation on large individuals will lead to a higher proportion of smaller individuals, while larger individuals reduce their metabolic activity. Without fish (bottom left), large invertebrates feeding on smaller individuals are predicted to lead to a higher proportion of larger individuals, while smaller individuals reduce their metabolism. The scatter plot (right) shows slopes of metabolic scaling (*b*, circle) and ISD slope (*λ*, triangle) under heated and unheated conditions, with green and purple points representing fish and no-fish treatments, respectively. Results indicate that *λ* remains stable (contrary to our hypotheses), while *b* varies with temperature and predation.

and vulnerability to predation, such species-level heterogeneity is expected to contribute background variability rather than drive systematic treatment effects in a continuously assembling community. The resilience of the ISD's λ observed across treatments may therefore reflect not only functional redundancy or trait compensation but also the stabilizing effect of ongoing immigration from regional species pools. These dynamics are rarely accounted for in closed experimental systems, where community responses can be more sensitive to perturbation. Our findings suggest that realistic colonization processes are crucial for understanding how ecosystems maintain structural integrity even as physiological responses vary at the individual level.

5. Conclusion

Overall, our findings highlight that λ is not solely determined by b. Instead, community-level structure appears to be maintained through trade-offs among key trophic parameters. Even when individual metabolic responses shift due to warming or predator presence, the broader energy distribution across body sizes may remain stable if underlying trophic efficiencies and interaction strengths adjust in tandem. Our findings reinforce the view that metabolic scaling is not a fixed, universal property, but one that is ecologically contingent—shaped by thermal conditions, species interactions and behavioural responses to predation. The observed shifts in metabolic scaling under different predator regimes, coupled with the stability of community structure, suggest that physiological and structural responses can be decoupled, particularly in dynamic, self-assembling communities. This challenges core assumptions of the MTE and underscores the need for models that incorporate biotic context, mortality risk and trophic interactions. Predicting ecological responses to climate warming will require moving beyond temperature in isolation and embracing the complexity of ecological networks and community assembly processes.

Ethics. This work did not require ethical approval from a human subject or animal welfare committee.

Data accessibility. All data and code are publicly available at [61].

Supplementary material is available online [62].

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. V.G.: conceptualization, data curation, formal analysis, investigation, methodology, resources, software, supervision, validation, visualization, writing—original draft, writing—review and editing; D.S.G.: conceptualization, writing—original draft, writing—review and editing; J.P.F.P.: conceptualization, methodology, writing—review and editing; J.R.J.: conceptualization, writing—review and editing; A.S.: data curation, resources; J.W.: data curation, resources; S.R.: data curation, software; T.W.: data curation, resources; J.S.W.: conceptualization, formal analysis, funding acquisition, methodology, resources, visualization, writing—original draft, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration. We declare we have no competing interests.

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